

EE 5239 Nonlinear Optimization

Lecture 9: Duality – Examples

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Outline

- 1 Duality for Linear Program
- 2 The KKT Condition
- 3 The SVM Problem and its Dual
- 4 Coordinate Descent for SVM

Discussion Problem: Primal-Dual LP

- Consider the following resource allocation problem

$$\begin{aligned} \max \quad & c_1x_1 + c_2x_2 + c_3x_3, && \text{(maximize profit)} \\ \text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \leq b_1, && \text{(raw material is limited)} \\ & a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2, && \text{(raw material is limited)} \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

- c_j profit per unit of product j produced
- b_i total raw material i
- a_{ij} units of raw material i needed to produce 1 unit of product j
- Easily written in the following matrix form

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}$$

Discussion Problem: Primal-Dual LP

- What is the dual problem for

$$\begin{array}{ll}\max_x & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0.\end{array}$$

- Vice versa?

Discussion Problem: Primal-Dual LP

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$$\begin{array}{ll}\max_x & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0.\end{array}$$

- Apply the duality result

- Vice versa?

Discussion Problem: Primal-Dual LP

- What is the dual problem for

$$\begin{aligned} \max_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}$$

- Apply the duality result
- The dual problem is

$$\begin{aligned} \min_y \quad & y^T b \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

- Vice versa?

Discussion Problem: Primal-Dual LP

- The primal problem is from the perspective of the firm
- Consider the following problem
- Suppose there is a buyer that wants to buy all the raw materials from the firm (b_1 and b_2 units)
- Use y_1 and y_2 to denote the price the buyer is willing to pay

Discussion Problem: Primal-Dual LP

- The primal problem is from the perspective of the firm
- Consider the following problem
- Suppose there is a buyer that wants to buy all the raw materials from the firm (b_1 and b_2 units)
- Use y_1 and y_2 to denote the price the buyer is willing to pay
- **Question:** How to setup an optimization problem that is beneficial to both the buyer and the firm?
 - Buyer wants to minimize the buying cost
 - Firm wants to make sure that its profit is not hurt

Discussion Problem: Primal-Dual LP

- If the factory produces 1 unit less of product j
 - Free up a_{1j} units of material 1
 - Free up a_{2j} units of material 2
- Buyer pays $a_{1j}y_1 + a_{2j}y_2$
- But shall the factory sell such material?
- Yes, only if what is sold to the buyer is more than what it can get from producing the actual product !

$$a_{1j}y_1 + a_{2j}y_2 \geq c_j, \quad j = 1, 2, 3.$$

- Then the buyer's optimization problem is given by

$$\begin{aligned} \min \quad & b_1y_1 + b_2y_2 \\ \text{s.t.} \quad & a_{11}y_1 + a_{21}y_2 \geq c_1 \\ & a_{12}y_1 + a_{22}y_2 \geq c_2 \\ & a_{13}y_1 + a_{23}y_2 \geq c_3 \\ & y_1, y_2 \geq 0 \end{aligned}$$

Summary

- Our previous discussion can be summarized as
-

$$\begin{array}{lll} \max_x & c^T x & \Leftrightarrow \min_y \quad y^T b \\ & Ax \leq b & \Leftrightarrow A^T y \geq c \\ & x \geq 0 & \Leftrightarrow y \geq 0 \end{array}$$

- Is this generally true?
- What can be said about these two problems? Relationships?

The KKT Condition

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & h_i(x) = 0, \quad i = 1, \dots, m \end{array} \quad (0.1)$$

$$g_j(x) \leq 0, \quad j = 1, \dots, r \quad (0.2)$$

Any optimal and dual pairs $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ must satisfy



Albert
Tucker

$$\nabla f(\tilde{x}) + \sum_{j=1}^r \tilde{\lambda}_j \nabla g_j(\tilde{x}) + \sum_{i=1}^m \tilde{\nu}_i \nabla h_i(\tilde{x}) = 0_{K \times 1},$$

$$g_j(\tilde{x}) \leq 0, \forall j = 1, \dots, r, \quad (\text{primal feasibility})$$

$$h_i(\tilde{x}) = 0, \forall i = 1, \dots, m, \quad (\text{primal feasibility})$$

$$\tilde{\lambda}_j \geq 0, \forall j = 1, \dots, r, \quad (\text{dual feasibility})$$

$$g_j(\tilde{x}) \times \tilde{\lambda}_j = 0, \forall j \quad (\text{complementarity}).$$



Harold
Kuhn

Observation

- The last requirement is called the complementarity condition

$$g_j(\tilde{x}) \times \tilde{\lambda}_j = 0$$

- This says if $g_j(\tilde{x})$ is not active (i.e. $= 0$), then the corresponding multiplier must be zero
- Make sense? This says that the non-active constraints do not have any role in the condition

$$\nabla f(\tilde{x}) + \sum_{j=1}^n \tilde{\lambda}_j \nabla g_j(\tilde{x}) + \sum_{i=1}^m \tilde{\nu}_i \nabla h_i(\tilde{x}) = 0_{K \times 1}$$

- KKT is a necessary condition for optimality (when regularity conditions such as Slater holds); becomes sufficient if the problem is convex, and Slater holds true

The simple case

- For the fun of it, just write down the simple KKT condition

$$\begin{aligned} \min (x - 5)^2, \quad \text{s.t. } 0 \leq x \leq 2. \\ \Rightarrow \min (x - 5)^2, \quad \text{s.t. } -x \leq 0, x - 2 \leq 0. \end{aligned}$$

- Introduce two dual variables y_1, y_2
- The following are our KKT conditions

$$\begin{aligned} 2(x - 5) - y_1 + y_2 &= 0 \\ y_1 \geq 0, y_2 &\geq 0 \\ -x \leq 0, x - 2 &\leq 0 \\ y_1 \times x = 0, (2 - x) \times y_2 &= 0. \end{aligned}$$

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The Dual of SVM Problem: Separable Case

- Let's apply the KKT condition for the SVM problem
- The problem we have was (separable case, absorbed x_0 into $\mathbf{x}$)

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{x}^T \mathbf{x} \\ \text{subject to} \quad & b_i(\mathbf{a}_i^T \mathbf{x}) \geq 1, \forall i \end{aligned} \tag{0.1}$$

- We have absorbed x_0 by letting $[\mathbf{a}_i, 1]^T$ be our new $\mathbf{a}_i$, and include one extra dimension in $\mathbf{x}$ as well

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- The Lagrangian

$$L(\mathbf{x}, \lambda) = \frac{1}{2} \|\mathbf{x}\|^2 - \sum_i \lambda_i b_i(\mathbf{a}_i^T \mathbf{x}) + \sum_i \lambda_i$$

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- Setting the gradient to 0 we get

$$\mathbf{x} = \sum_i \lambda_i b_i \mathbf{a}_i$$

The Dual of SVM Problem: Separable Case

- Substituting these back, we get the Lagrangian dual

$$L^*(\lambda) = \sum_i \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j b_i b_j \mathbf{a}_i^T \mathbf{a}_j$$

- Constraints of the dual optimization

$$\lambda_i \geq 0, \forall i$$

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- So we can obtain the optimal dual variables $\{\lambda_i\}$ by solving

$$\begin{aligned} \max_{\lambda} L^*(\lambda) &= \sum_i \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j b_i b_j \mathbf{a}_i^T \mathbf{a}_j \\ \text{subject to } &\lambda_i \geq 0, \forall i \end{aligned}$$

- Quadratic objective, simple constraints!

The Dual of SVM Problem: Separable Case

- Once we solved the λ -problem, how to construct the model $\mathbf{x}$?
- This means

$$\lambda_i > 0 \rightarrow b_i(\mathbf{a}_i\mathbf{x}) = 1, \quad \mathbf{a}_i \text{ is a support vector} \quad (0.2)$$

$$\lambda_i = 0 \rightarrow b_i(\mathbf{a}_i\mathbf{x}) > 1, \quad \mathbf{a}_i \text{ is not a support vector} \quad (0.3)$$

The Dual of SVM Problem: Separable Case

- Once we solved the λ -problem, how to construct the model $\mathbf{x}$?
- Recall the complementary slackness $\lambda_i g_i(\mathbf{x}) = 0$ for $g_i(\mathbf{x}) \leq 0$
- This means

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- The resulting $\mathbf{x}$ is given by (make sense?)

$$\mathbf{x} = \sum_{i: \lambda_i > 0} \lambda_i b_i \mathbf{a}_i$$

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- Optimal model is a **nonnegative combination** of the data points!

The Dual of SVM Problem: Non-Separable Case

- The problem for non-separable case, (absorbed x_0 into $\mathbf{x}$)

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{x}^T \mathbf{x} + c \sum_i \xi_i \\ \text{subject to} \quad & b_i(\mathbf{a}_i^T \mathbf{x}) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i. \end{aligned} \quad (0.4)$$

- Let's form the Lagrangian, setup the KKT and solve it

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- The Lagrangian (Let $\mu_i \geq 0$ denote the multiplier for the constraint $\xi_i \geq 0$)

$$\frac{1}{2} \|\mathbf{x}\|^2 + c \sum_i \xi_i - \sum_i \lambda_i (b_i(\mathbf{a}_i^T \mathbf{x}) - 1 + \xi_i) - \sum_i \mu_i \xi_i$$

The Dual of SVM Problem: Non-Separable Case

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- Setting gradient w.r.t. $(\mathbf{x}, \xi)$ to zero

$$\mathbf{x} = \sum_i \lambda_i b_i \mathbf{a}_i, \quad 0 \leq \lambda_i \leq c, \quad \mu_i = c - \lambda_i$$

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- Setting gradient w.r.t. $(\mathbf{x}, \xi)$ to zero

$$\mathbf{x} = \sum_i \lambda_i b_i \mathbf{a}_i, \quad 0 \leq \lambda_i \leq c, \quad \mu_i = c - \lambda_i$$

- If we know the optimal λ_i , then we can construct $\mathbf{x}$

The Dual of SVM Problem: Non-Separable Case

- Substituting, we get the Lagrangian dual

$$L^*(\lambda) = \sum_i \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j b_i b_j \mathbf{a}_i^T \mathbf{a}_j$$

- Constraints of the dual optimization

$$0 \leq \lambda_i \leq c, \quad \forall i$$

- Recall the complementary slackness $\lambda_i g_i(\mathbf{x}) = 0$ for $g_i(x) \leq 0$
- This means

$$\lambda_i > 0 \rightarrow b_i(\mathbf{a}_i \mathbf{x}) + \xi_i = 1, \quad \mathbf{a}_i \text{ is a support vector} \quad (0.5)$$

$$\lambda_i = 0 \rightarrow b_i(\mathbf{a}_i \mathbf{x}) + \xi_i > 1, \quad \mathbf{a}_i \text{ is not a support vector} \quad (0.6)$$

- How to obtain ξ_i ? By primal feasibility. If $\mathbf{x}$ is known, then it is easy to determine based on:

$$b_i(\mathbf{a}_i^T \mathbf{x}) - 1 + \xi_i \geq 0, \quad \xi_i \geq 0$$

The Dual of SVM Problem: summary

- Step 1: Solve the following problem

$$\begin{aligned} \max_{\lambda} L^*(\lambda) &= \sum_i \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j b_i b_j \mathbf{a}_i^T \mathbf{a}_j \\ \text{subject to} \quad &0 \leq \lambda_i \leq c, \quad \forall i \end{aligned}$$

- Step 2: Construct the model $\mathbf{x}$ by

$$\mathbf{x} = \sum_i \lambda_i b_i \mathbf{a}_i.$$

- Step 3: Solve for ξ_i

$$b_i(\mathbf{a}_i^T \mathbf{x}) - 1 + \xi_i \geq 0, \quad \xi_i \geq 0$$

The last step is not really useful for building the model, but tells you the “training errors”

Summary

- For both cases, the original “difficult” problem, when viewed in a dual perspective, becomes a simple quadratic problem with **bound** constraints

$$\begin{aligned} \max \quad & \sum_i \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j b_i b_j \mathbf{a}_i^T \mathbf{a}_j \\ \text{subject to.} \quad & 0 \leq \lambda_i \leq c, \forall i \end{aligned}$$

- How can we solve it? Any candidate methods?

A Popular Method: CD on the Dual

- A very popular method ¹ for linear SVM is coordinate descent (CD)
- Fixing all dual variables, update **one** λ_i at a time
- You will see that by doing this, each time we are effectively processing **one data point** at a time

A Popular Method: CD on the Dual

- Gradient w.r.t. λ_i
(one dual coordinate $\iff$ one primal data point)

$$\nabla_i L^*(\lambda^r) = 1 - b_i \mathbf{a}_i^T \sum_j \lambda_j^r b_j \mathbf{a}_j = 1 - b_i \mathbf{a}_i^T \mathbf{x}^t$$

- Perform gradient projection on λ_i (why this specific stepsize?)

$$\lambda_i^{r+1} = \text{proj} \left[\lambda_i^r + \frac{1}{b_i^2 \|\mathbf{a}_i^T\|^2} \nabla_i L^*(\lambda^r) \right], \text{ if coordinate } i \text{ is picked}$$

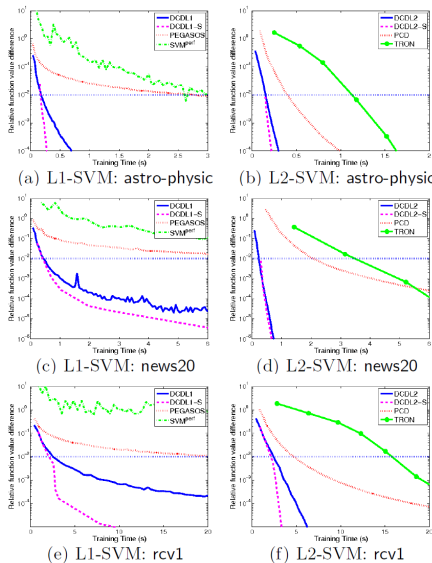
(Note we are maximizing, so walk towards gradient direction)

- $\text{proj}[x]$ means projection to $0 \leq x \leq c$; simple solution

$$\max\{\min\{x, c\}, 0\}$$

- **Key:** Pick one data point at a time, improve the classifier gradually

Performance



Conclusion

- We see how to use KKT solution to characterize the (sub)optimal primal-dual pairs
- We see how this useful concept are applied to linear SVM
 - ① Convert a problem with a set of difficult constraints to simple ones
 - ② For both separable and unseparable problems
 - ③ Design simple (yet very effective) algorithms to solve the problem